Geometry-Family Attribution Extension

Independent geometry replays, cross-seed controls, and longer depth/strength controls for the InstructPix2Pix pipeline

20-iteration family runs	Cases	Cross-seed replay seeds	Long controls
36	2	1234, 24001, 34007	10

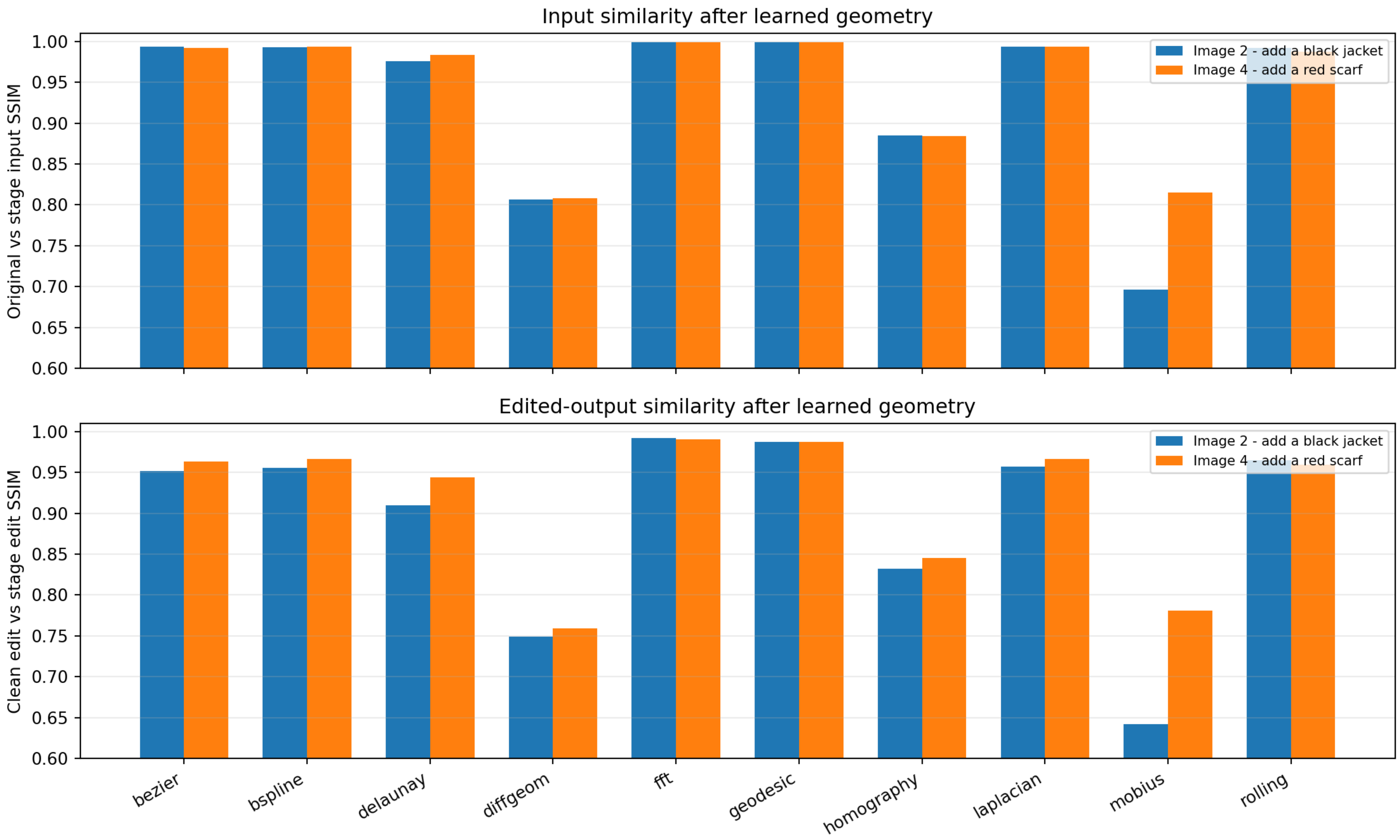
Extension finding

The broad family sweep confirms that geometry can change edited outputs, but the largest single-family changes usually coincide with visible input deformation. B-spline is mild on the original image. At the fixed seed, longer B-spline or rolling optimization after VAE reconstruction can cross an editor discontinuity and produce a severe jacket collapse; the cross-seed controls determine whether that behavior is repeatable.

The Image 4 red-scarf clean baseline does not form a convincing scarf and is retained only as a sensitivity control, not as an attack-success case.

A1. Single-family geometry effects

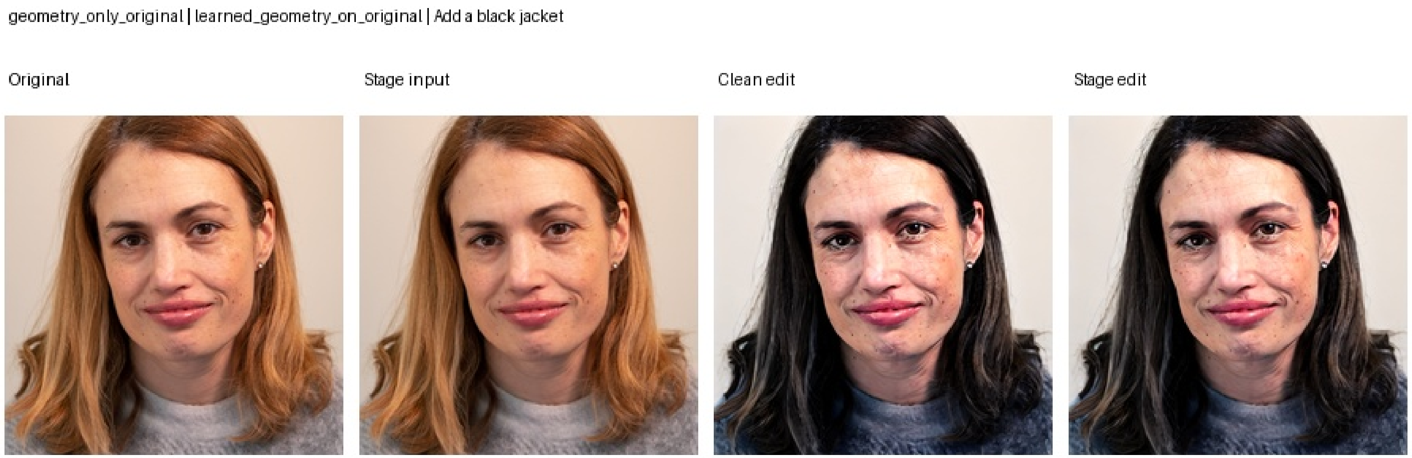
Each point uses the independently optimized geometry-only-on-original branch. Lower edited-output SSIM means a larger pixel-level edit change, but it must be read together with input SSIM and the saved images.



A2. Image 2 - add a black jacket

Family	Input SSIM	Edit SSIM vs clean	Edit ArcFace vs clean	Input PSNR
bezier	0.994	0.951	0.991	44.26
bspline	0.993	0.955	0.992	44.16
delaunay	0.976	0.910	0.963	37.92
diffgeom	0.807	0.749	0.881	27.00
fft	0.999	0.992	0.994	49.97
geodesic	0.999	0.988	0.994	55.88
homography	0.885	0.832	0.938	31.52
laplacian	0.994	0.957	0.991	44.11
mobius	0.696	0.642	0.716	25.02
rolling	0.992	0.965	0.993	43.48

B-spline geometry-only replay



Differential-geometry replay

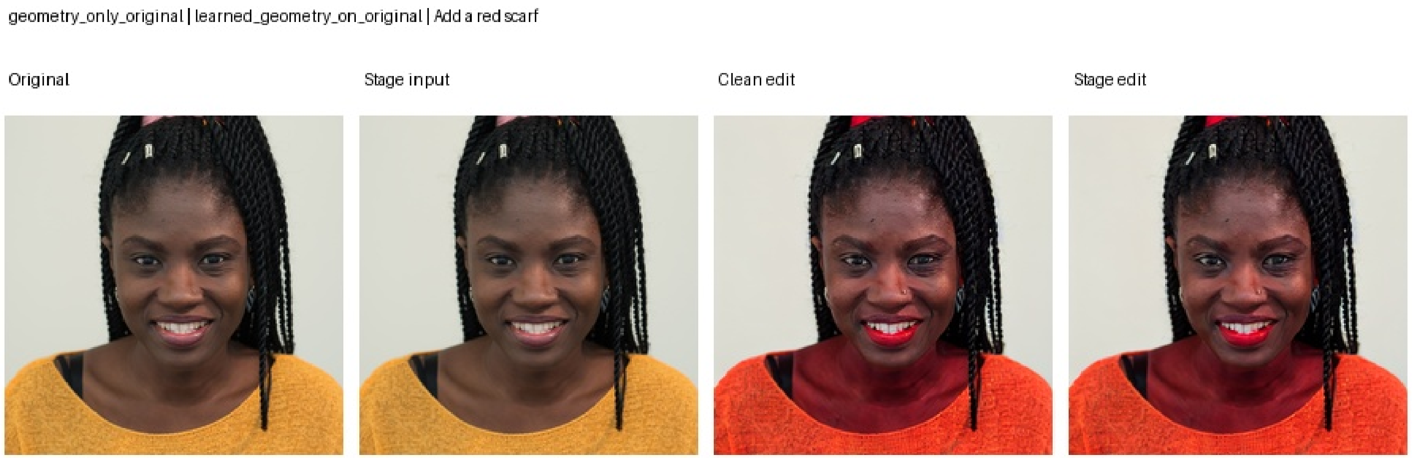
geometry_only_original | learned_geometry_on_original | Add a black jacket

Original | Stage input | Clean edit | Stage edit

A2. Image 4 - add a red scarf

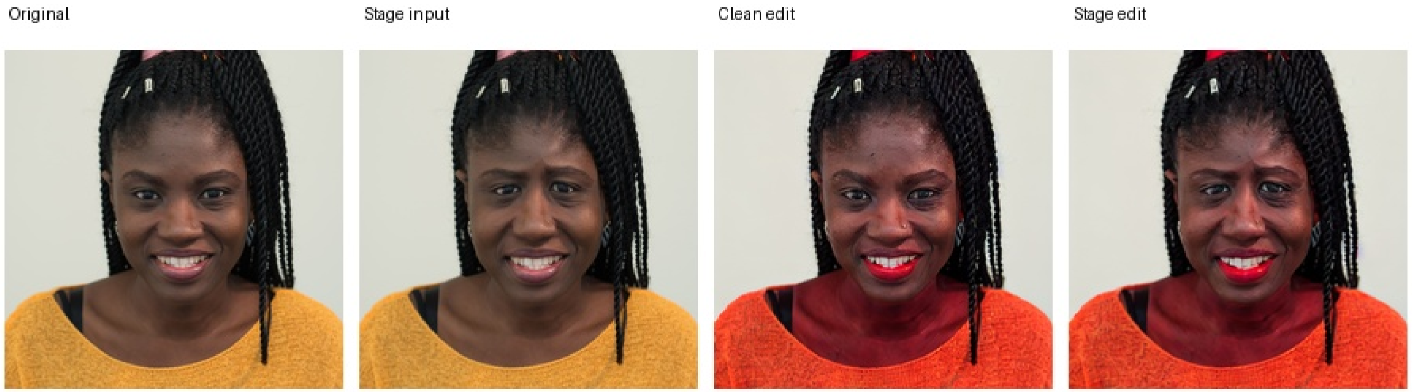
Family	Input SSIM	Edit SSIM vs clean	Edit ArcFace vs clean	Input PSNR
bezier	0.992	0.964	0.974	41.87
bspline	0.993	0.966	0.986	42.83
delaunay	0.984	0.944	0.975	39.38
diffgeom	0.808	0.759	0.674	23.41
fft	0.999	0.990	0.997	50.51
geodesic	0.999	0.988	0.996	56.88
homography	0.884	0.845	0.934	28.30
laplacian	0.994	0.967	0.984	42.21
mobius	0.815	0.781	0.917	26.16
rolling	0.987	0.959	0.991	37.47

B-spline geometry-only replay



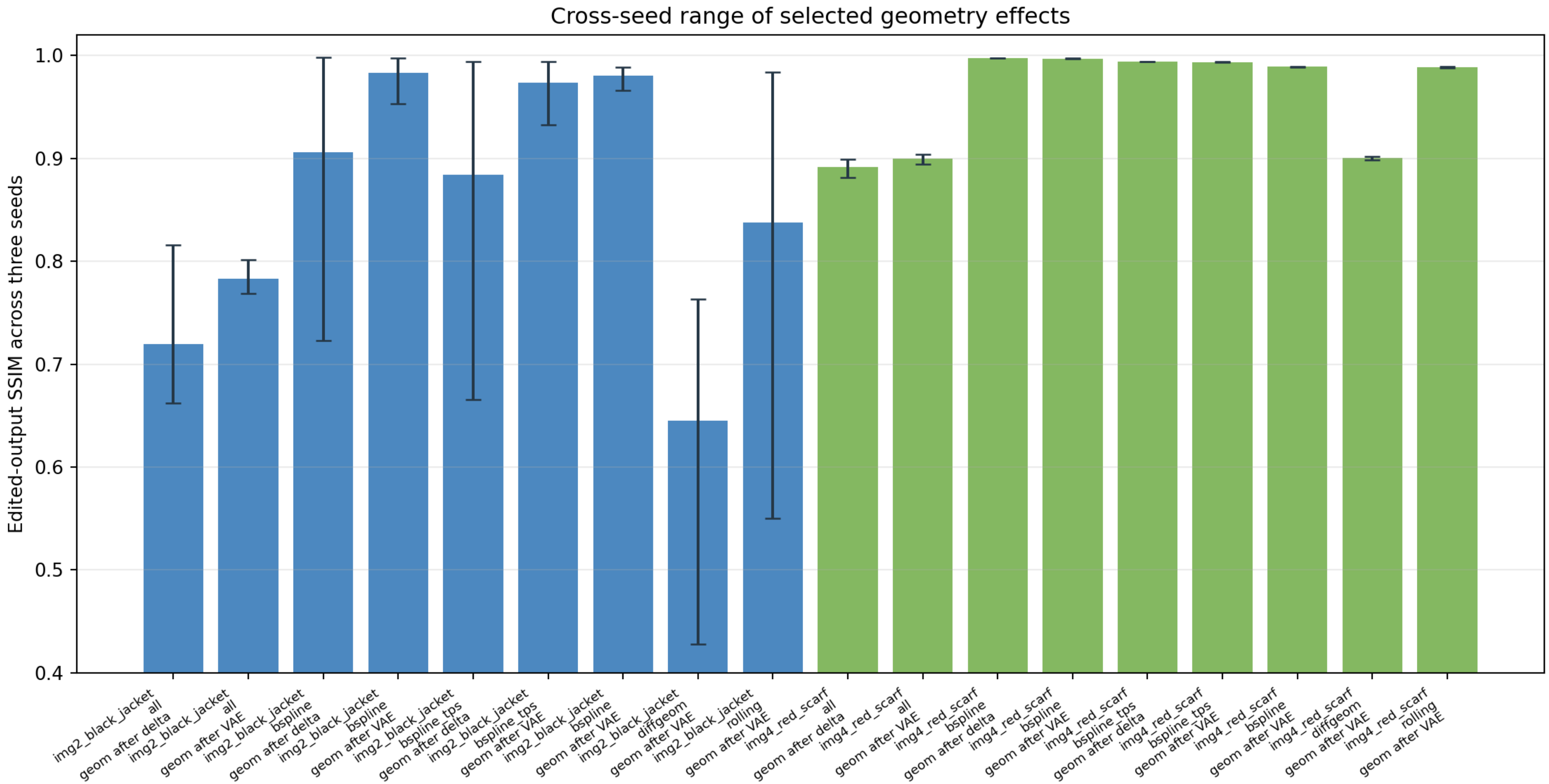
Differential-geometry replay

geometry_only_original | learned_geometry_on_original | Add a red scarf



A3. Cross-seed stability

Bars show mean edited-output SSIM and error bars show the observed min-to-max range across three deterministic diffusion seeds. A wide range indicates editor-seed sensitivity rather than a stable transformation effect.



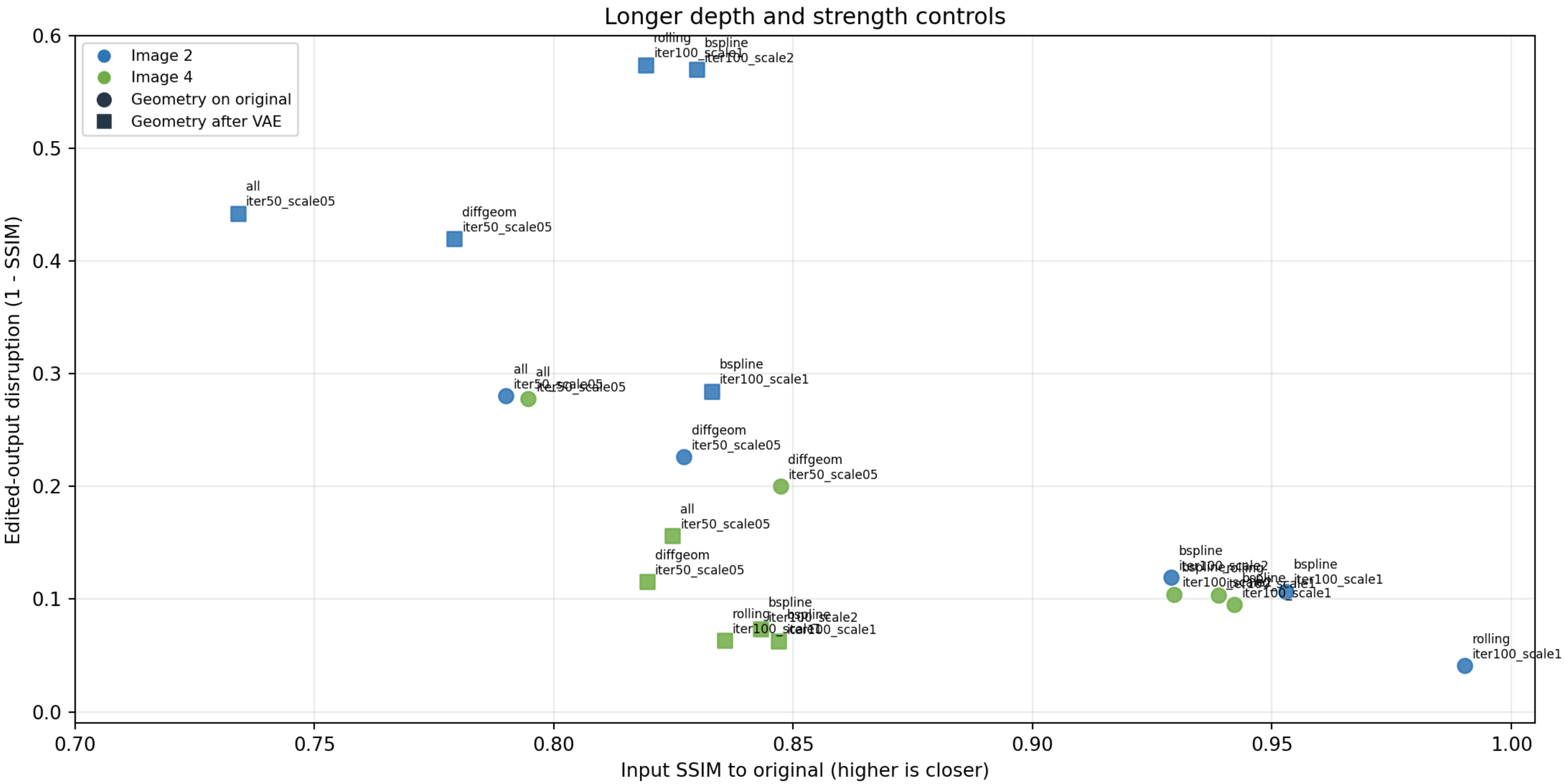
A3.1 Image 2 cross-seed metrics

Run	Comparison	Mean SSIM	Min	Max	Range
combined_bspline	geometry increment after delta	0.906	0.723	0.998	0.275
combined_bspline	learned geometry vs vae reconstruction	0.983	0.953	0.998	0.044
combined_bspline_tps	geometry increment after delta	0.884	0.665	0.994	0.329
combined_bspline_tps	learned geometry vs vae reconstruction	0.973	0.932	0.994	0.062
single_bspline	learned geometry vs vae reconstruction	0.980	0.966	0.989	0.023
single_diffgeom	learned geometry vs vae reconstruction	0.645	0.428	0.763	0.335
single_rolling	learned geometry vs vae reconstruction	0.838	0.550	0.984	0.433

At 20 iterations, B-spline on the reconstructed input is close to its neutral baseline for seeds 1234 and 34007 but changes sharply at seed 24001. Differential geometry is stronger across seeds, but it also produces visible facial deformation in the stage input.

A4. Longer depth and strength controls

B-spline and rolling were extended to 100 iterations. Differential geometry and the all-family combination were tested at reduced amplitude. The plot separates input similarity from edited-output disruption and distinguishes geometry applied to the original from geometry applied after VAE reconstruction.



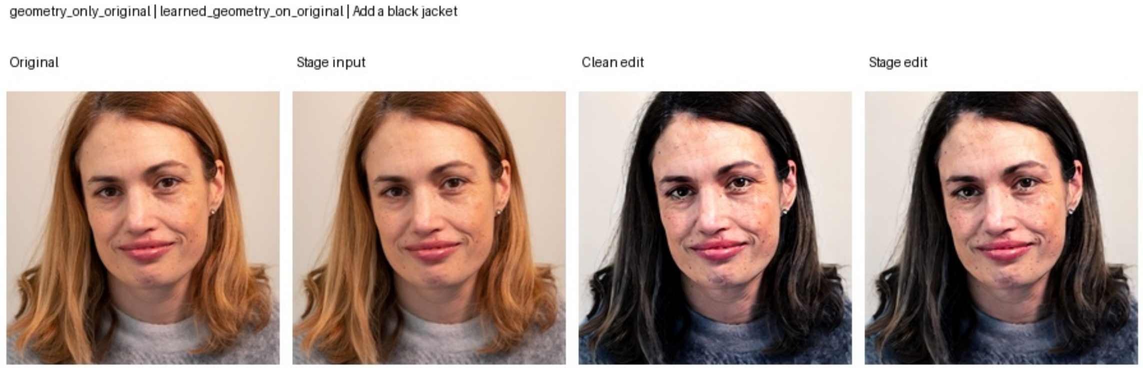
A4.1 Direct long-control metrics

Case	Control	Branch	Edit SSIM	ArcFace edit similarity
image2 black jacket	all_iter50_scale05	original	0.720	0.699
image2 black jacket	all_iter50_scale05	reconstruction	0.544	0.359
image2 black jacket	bspline_iter100_scale1	original	0.893	0.929
image2 black jacket	bspline_iter100_scale1	reconstruction	0.772	0.736
image2 black jacket	bspline_iter100_scale2	original	0.881	0.911
image2 black jacket	bspline_iter100_scale2	reconstruction	0.467	0.309
image2 black jacket	diffgeom_iter50_scale05	original	0.774	0.832
image2 black jacket	diffgeom_iter50_scale05	reconstruction	0.612	0.602
image2 black jacket	rolling_iter100_scale1	original	0.959	0.989
image2 black jacket	rolling_iter100_scale1	reconstruction	0.464	0.350
image4 red scarf	all_iter50_scale05	original	0.722	0.402
image4 red scarf	all_iter50_scale05	reconstruction	0.857	0.867
image4 red scarf	bspline_iter100_scale1	original	0.905	0.940
image4 red scarf	bspline_iter100_scale1	reconstruction	0.973	0.985
image4 red scarf	bspline_iter100_scale2	original	0.896	0.926
image4 red scarf	bspline_iter100_scale2	reconstruction	0.959	0.976
image4 red scarf	diffgeom_iter50_scale05	original	0.800	0.636
image4 red scarf	diffgeom_iter50_scale05	reconstruction	0.910	0.901
image4 red scarf	rolling_iter100_scale1	original	0.896	0.914
image4 red scarf	rolling_iter100_scale1	reconstruction	0.979	0.993

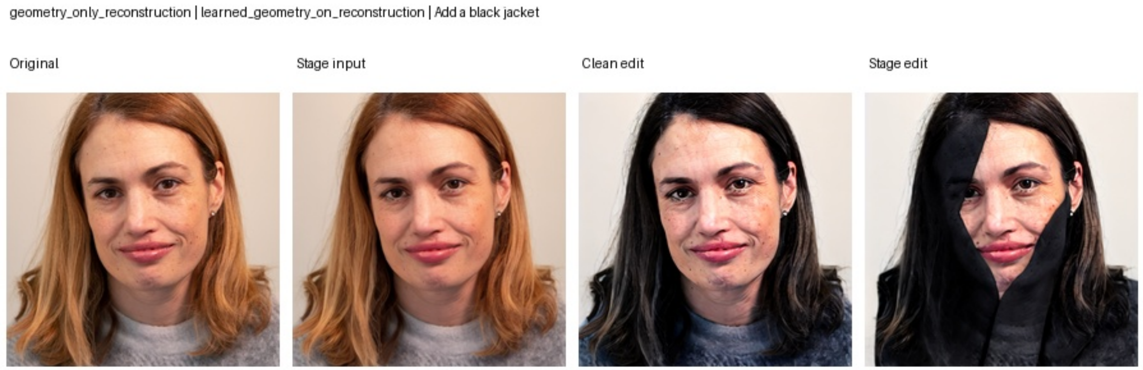
The 100-iteration B-spline branch remains visually close to the clean edit when applied directly to the original. After VAE reconstruction, scale 1 produces a hood-like jacket at seed 24001 and scale 2 produces a stronger full-jacket collapse. Rolling shows the same stage interaction at this seed.

A4.2 Image 2 long-control images

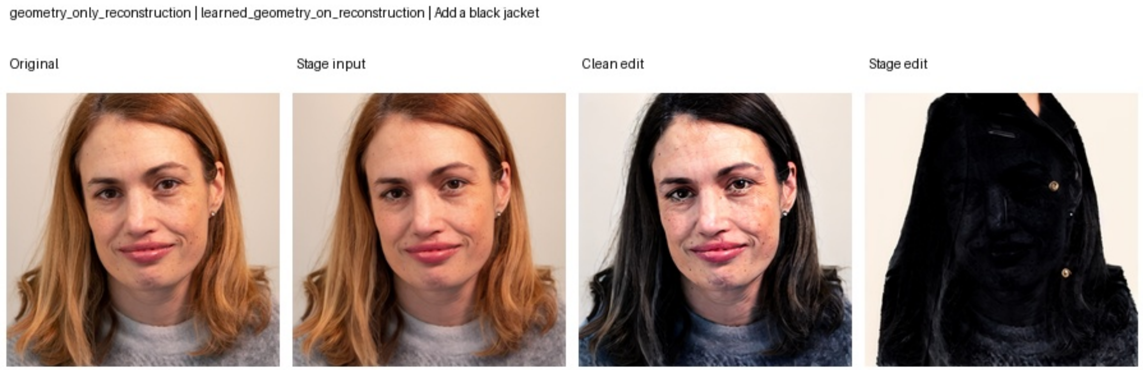
B-spline, 100 iterations, scale 1 - geometry on original



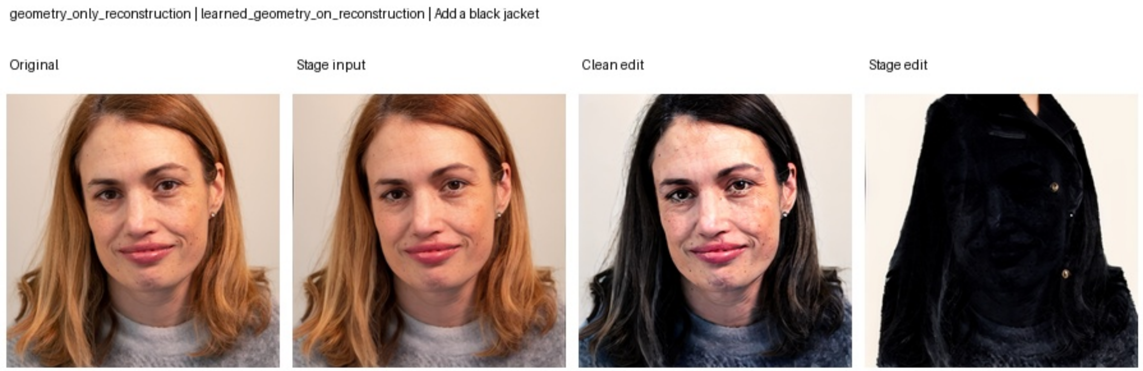
B-spline, 100 iterations, scale 1 - geometry after VAE



B-spline, 100 iterations, scale 2 - geometry after VAE



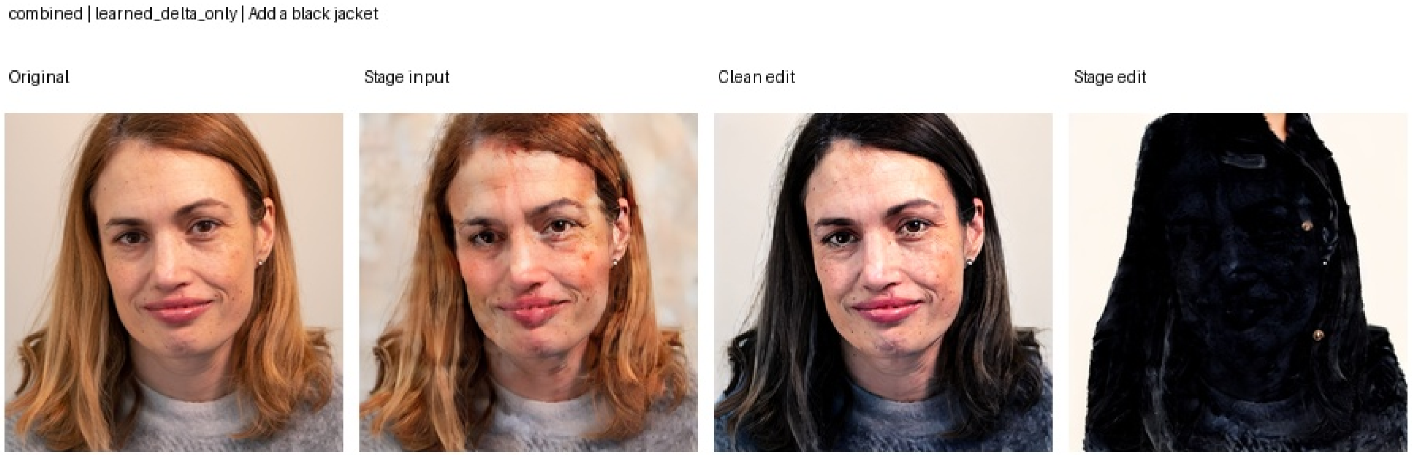
Rolling, 100 iterations - geometry after VAE



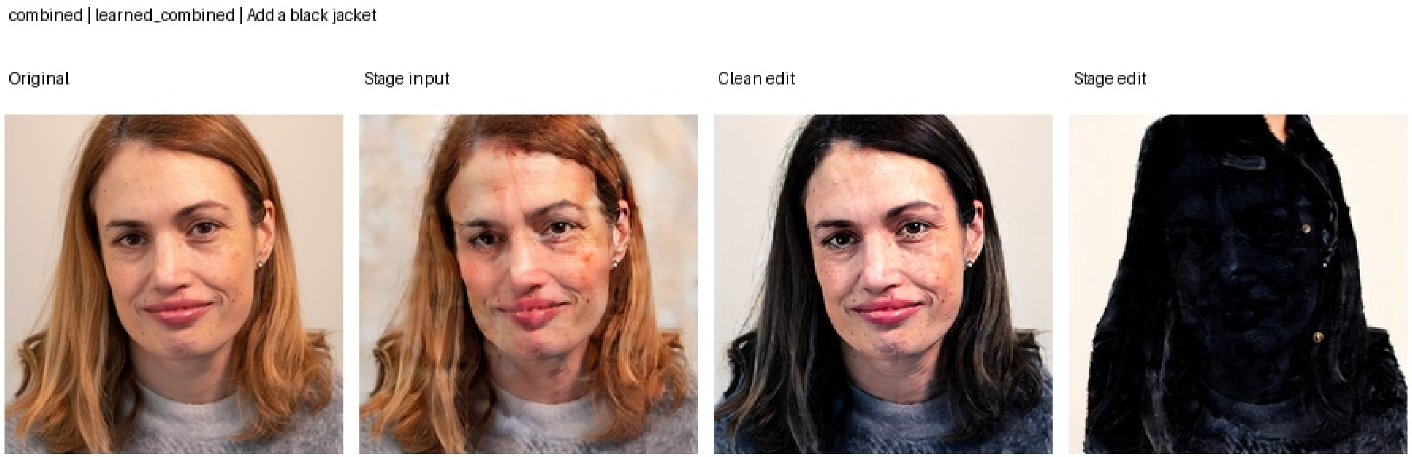
A4.3 Combined-path attribution

The joint run preserves the earlier attribution: delta-z already produces the severe collapse, and adding B-spline changes the saved edit only slightly (geometry-after-delta SSIM 0.991 at scale 1 and 0.990 at scale 2).

Latent delta-z only within the B-spline joint run

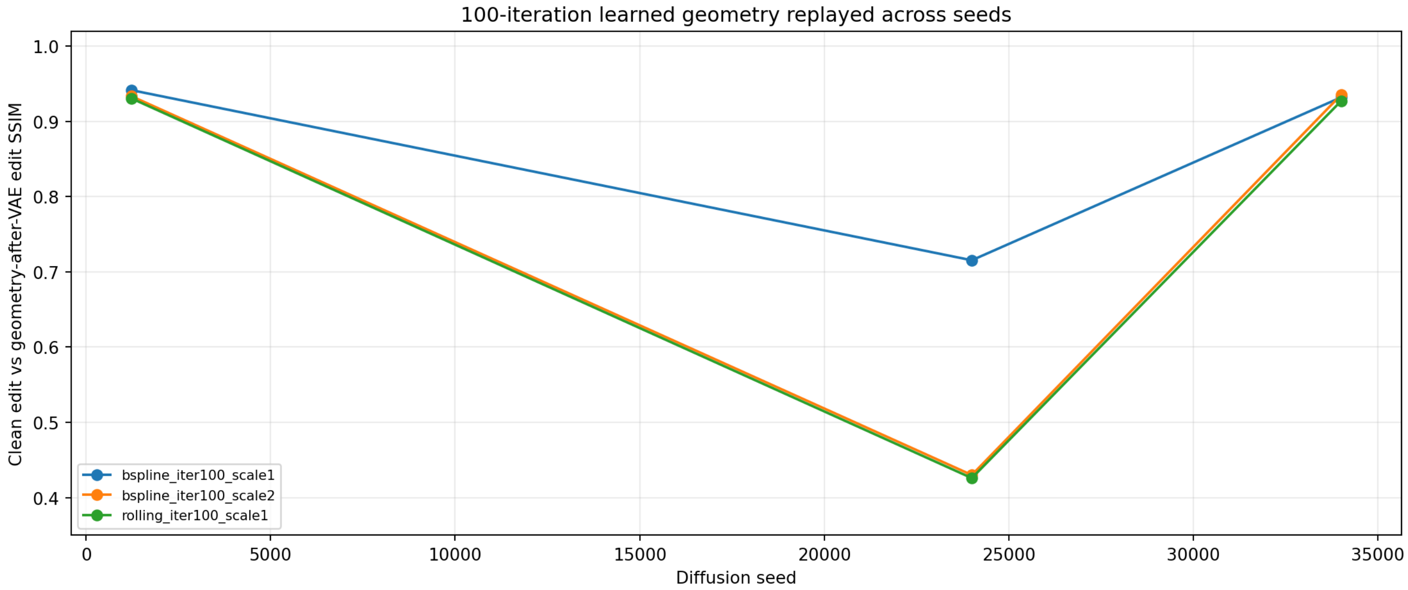


Latent delta-z plus B-spline



A5. 100-iteration geometry seed replay

The exact saved long-control inputs were re-edited at three seeds without rerunning optimization. B-spline scale 2 produces output SSIM 0.934, 0.430, and 0.935 at seeds 1234, 24001, and 34007. Rolling produces 0.931, 0.426, and 0.927. The collapse is therefore concentrated at seed 24001 rather than stable across seeds.

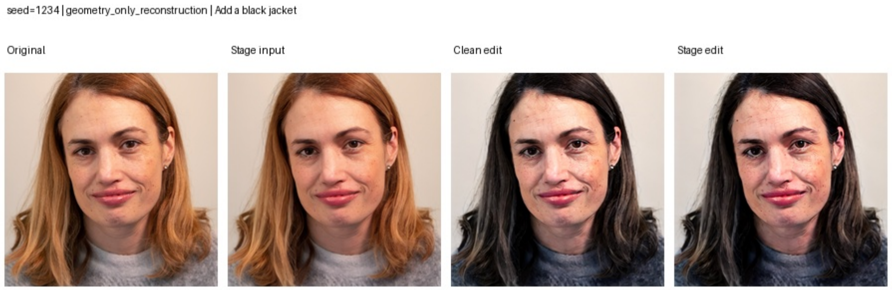


Control	Seed	Stage	Input SSIM	Output SSIM	Edit ArcFace
bspline_iter100_scale1	1234	geometry after VAE	0.833	0.942	0.971
bspline_iter100_scale1	24001	geometry after VAE	0.833	0.715	0.721
bspline_iter100_scale1	34007	geometry after VAE	0.833	0.932	0.922
bspline_iter100_scale2	1234	geometry after VAE	0.830	0.934	0.963
bspline_iter100_scale2	24001	geometry after VAE	0.830	0.430	0.321
bspline_iter100_scale2	34007	geometry after VAE	0.830	0.935	0.933
rolling_iter100_scale1	1234	geometry after VAE	0.819	0.931	0.982
rolling_iter100_scale1	24001	geometry after VAE	0.819	0.426	0.375
rolling_iter100_scale1	34007	geometry after VAE	0.819	0.927	0.946

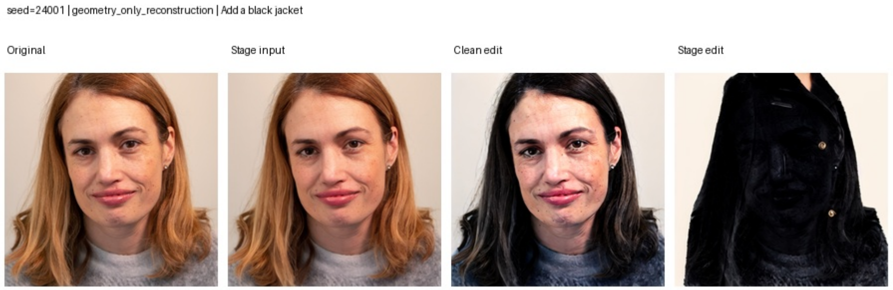
A5.1 Exact B-spline scale-2 input across seeds

The stage input is identical in all three rows. Only the diffusion seed changes. The saved strips visually confirm that the severe jacket collapse occurs at seed 24001 and not at the other two seeds.

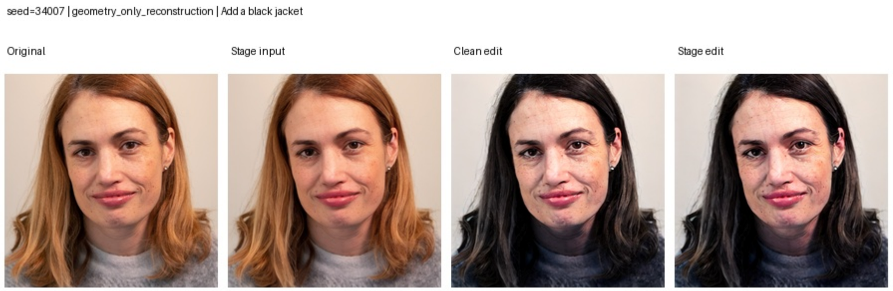
Seed 1234



Seed 24001



Seed 34007



A6. Updated attribution

1. The original report's central finding is unchanged: optimized latent delta-z is the dominant learned source in the current combined pipeline.

2. B-spline is mild when optimized and replayed directly on the original image, including at 100 iterations and doubled configured scale.

3. Differential geometry, Mobius, and homography can move edited outputs more strongly, but the saved inputs are visibly deformed; these are not subtle geometry-only effects.

4. VAE reconstruction can place a mild learned warp near an InstructPix2Pix decision boundary. Long B-spline and rolling controls collapse the jacket edit at seed 24001, while the exact same inputs produce normal edits at seeds 1234 and 34007.

5. The geometry-after-delta increment remains small in the long B-spline joint controls. The joint-path collapse is already present in the delta-z-only replay.

6. Image 4 confirms geometric sensitivity but cannot establish edit failure because its clean red-scarf baseline is semantically weak.

7. SSIM and ArcFace values are used as diagnostics and are interpreted together with the saved image strips; neither metric alone determines semantic edit success.

Final conclusion: the observed InstructPix2Pix changes are primarily latent-delta effects, with a separate seed-sensitive interaction between VAE reconstruction and learned geometry. The evidence does not support geometry alone as a stable, subtle cause of the strongest failures.